

Groups and Lie Algebras: Warm-up Sheet

1. **Tensor products and direct sums of linear maps.** Recall, or look up, the definition of the tensor product and direct sum of vector spaces.

- a) Let V be a 2-dimensional complex vector space and e_1 and e_2 a basis of V . Consider the two matrices A_1 and A_2 in that basis:

$$A_1 = \begin{pmatrix} e^{2\pi i/3} & 0 \\ 0 & e^{-2\pi i/3} \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Describe the set of all matrices that can be written as products of arbitrarily many copies of A_1 and A_2 .

- b) Compute the matrices $B_i = A_i \oplus A_i$ and $C_i = A_i \otimes A_i$ for $i = 1, 2$.
- c) What is the size of the set of 4×4 matrices that can be written as a product of arbitrarily many copies of B_1 and B_2 . Answer the same question for C_1 and C_2 . (You don't have to do any additional matrix multiplications).

2. **Examples of groups.** The structure we have seen in the previous exercise is called a *group*. It is a set G equipped with an associative and unital binary operation such that every element $g \in G$ admits an inverse.

- a) What are some examples of groups you have encountered before?
- b) Why do symmetries form a group? What are the groups of symmetries of the square and circle?

3. **Direct product.** Consider two groups G_1 and G_2 . Show that the set of ordered pairs (g_1, g_2) , where $g_1 \in G_1$, $g_2 \in G_2$, together with the multiplication law

$$(g_1, g_2) \cdot (h_1, h_2) = (g_1 \cdot h_1, g_2 \cdot h_2)$$

forms a group. It is called the direct product $G_1 \times G_2$.

4. **Reflection symmetry in quantum mechanics.** Denote by $\mathcal{H} = L^2(\mathbb{R})$ the Hilbert space of square-integrable functions on the real line. Consider the operator

$$S: \mathcal{H} \longrightarrow \mathcal{H} \\ f(x) \longmapsto f(-x) \quad .$$

- a) Show that the eigenvalues of S are ± 1 and for every function f , construct $+1$ and -1 eigenfunctions f_+ and f_- such that $f = f_+ + f_-$.
- b) Let $\mathbb{H}: \mathcal{H} \rightarrow \mathcal{H}$ be a Hamiltonian that commutes with S . Show that the transition amplitudes $\langle f_- | \mathbb{H} | f_+ \rangle$ vanish for all f_+ and f_- that satisfy $Sf_+ = f_+$ and $Sf_- = -f_-$.